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SUPERSEDING
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24 May 1972

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5847A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Pentode, miniature, high gain

Outline	---	6-1 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	g1	nc	h, lower int sd	k, g3 upper int sd	nc	a	nc	g2	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Rk	Ik	Pp	Pg2	Ehk	TE	Alt
Unit:	V	V dc	V dc	V dc	Ohms	mA dc	W	W	v	°C	ft
Maximum:	6.9	200	---	165	---	40	3.3	0.85	100	155	Note 1
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	150	0	150	110	---	---	---	---	---	---

GENERAL:

Qualification - Required (see note 2)

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1	3					
1266	Total grid current	4,5		Ic1	0	-0.2	µA dc
1256	Electrode current (1) (anode)			Ib	9.0	19.0	mA dc
1256	Electrode current (2) (anode)		Ec1 = -10 V dc	Ib	---	50	µA dc
1256	Electrode current (screen)			Ic2	---	6.0	mA dc
1306	Transconductance (1)			Sm	10,000	18,000	µmhos
1306	Transconductance (2)		Ef = 5.7 V	$\frac{\Delta Sm}{Ef}$	---	15	%
1306	Transconductance (3)		Eb = Ec2 = 150 V; Rk = 4,000 ohms; Ec1 = +2.5 V dc	Sm	2,600	3,700	µmhos
1306	Transconductance (4)		Eb = Ec2 = 150 V; Rk = 4,000 ohms; Ec1 = +20 V dc	Sm	6,500	9,450	µmhos
1201	Short and discontinuity detection			---	---	---	---
	Quality conformance inspection, part 2						
1031	Low-frequency vibration		Rp = 2,000 ohms	Ep	---	500	mV ac
1301	Heater current			If	280	320	mA
1336	Heater-cathode leakage			Ihk	---	20	µA dc
1331	Direct-interelectrode capacitance		Shield no. 315	Cgp	---	0.06	pF
				Cin	7.0	7.7	pF
				Cout	2.6	3.3	pF
1211	Insulation of electrodes			R	200	---	Meg
1121	Base strain			---	---	---	---
2126	Envelope strain	6		---	---	---	---
1105	Permanence of marking			---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1501	Intermittent life		Group A; Ehk = 100 V	---	---	---	---
	Intermittent life-test end points (500 hours):						
1306	Transconductance (1)			Sm	8,500	---	µmhos
1266	Total grid current	4		Ic1	---	-0.25	µA dc
1306	Transconductance (3)		Eb = Ec2 = 150 V; Rk = 4,000 ohms; Ec1 = +2.5 V dc	ΔSm t	---	-5	%
1211	Insulation of electrodes			R	50	---	Meg

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. In addition to the qualification requirements specified, tubes shall perform satisfactorily in Nike-Hercules Intermediate-Frequency (IF) Amplifier GS-19787. Twenty-five tubes will be furnished to the qualification approval authority for this test.
3. The AQL shall be 0.65, inspection level II, except for method 1201 (short and discontinuity detection), where the AQL shall be 0.4, inspection level II.
4. The following test is approved as an alternate for this test. The insertion of a 0.47-megohm resistor in the grid circuit shall not change the anode current by more than 0.5 mA dc.
5. This test shall be performed at the conclusion of the holding period.
6. The AQL shall be 2.5, inspection level I.
7. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 99
DLA - ES

User activities:
Army - AR, MI
Navy - AS, OS, MC, CG,
Air Force - 11, 14, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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